

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 6. Loops - Control Flow

Lecture: 52. Logic using While Loop

Section 1: Lecture Summary

This lecture focuses on practical usage of the **while loop** in Python through example programs. It emphasizes understanding the counter concept as a variable that increments with each loop iteration, which is key in controlling loops. The lecture first demonstrates a counting loop that prints multiples of a number (e.g., multiplication table for 3). Next, it explains how to extract digits from an integer using the modulus and floor division operations with while loops. The lecturer traces program execution step-by-step to solidify understanding of how variables change inside loops, especially for appreciating loop behavior and debugging. The overall goal is to build a strong foundational logic for writing and tracing loops for solving programming problems.

Section 2: Key Concepts and Explanations

- While Loop and Counter Logic

- A **counter** is a variable used to keep track of iterations, often initialized to zero.
- `i = i + 1` increments the counter by one on each loop iteration.
- The while loop condition `while i < 10:` means the loop runs as long as `i` is less than 10.
- Inside the loop, the value of `i` updates and controls when the loop terminates.
- Typical usage: To repeat a block of code a specific number of times.

- Multiplication Table Program Logic Using While Loop

- Initialize counter `i = 0`.
- Set a number `n`, e.g., 3.
- Inside the loop:
 - Increment `i` by one ($i = i + 1$).
 - Print the product of `n` and `i`.
- Loop stops when `i` reaches 10.
- Output is the multiplication table of 3 up to 3×10 .

- **Extracting Digits from a Number**

- Use modulus operator `% 10` to get the last digit (remainder after division by 10).
- Use floor division `/ / 10` to remove the last digit from the number.
- Repeat this inside a while loop as long as the number is greater than zero.
- This process extracts each digit from right to left.

- **Tracing Loops**

- Tracing involves writing down variable values as they change line by line.
- Use tables to record changing values of key variables in the loop.
- Helps understand and debug the loop's logic and behavior.

Section 3: Example Code and Use Cases

1. Counting Loop to Print Multiplication Table for 3

```
n = 3          # Number for multiplication table
i = 0          # Counter initialized to zero

while i < 10:
    i = i + 1    # Increment counter by 1
    print(n * i) # Print product of n and i
```

Explanation: This program prints multiples of 3 from 3 up to 30, incrementing the multiplier from 1 to 10.

2. Enhanced Multiplication Table with Format

```
n = 3
i = 0

while i < 10:
    i = i + 1
    print(n, 'x', i, '=', n * i)
```

Explanation: Prints a formatted multiplication table like "3 x 1 = 3", showing both factors and product clearly.

3. Extracting Digits from a Number

```
n = 4365

while n > 0:
    r = n % 10    # Get last digit
```

```
print(r)          # Print last digit
n = n // 10       # Remove last digit from number
```

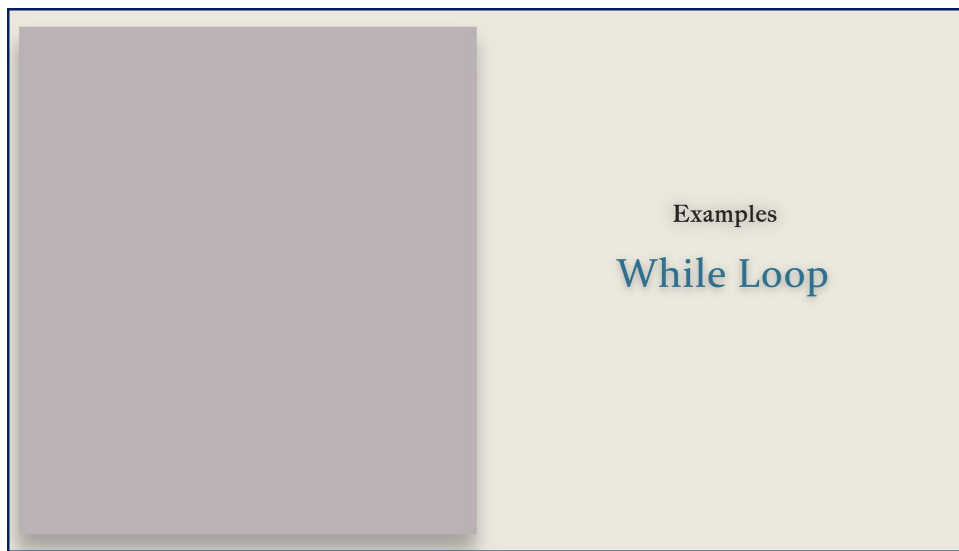
Explanation: This program prints each digit of the number 4365 from right to left (5, then 6, then 3, then 4), by repeatedly extracting and discarding the last digit.

Section 4: Key Takeaways

- The **counter variable** (commonly `i`) is fundamental for controlling the number of loop iterations.
- `i = i + 1` is the incremental step that updates the counter inside loops.
- The **while loop** executes as long as its condition remains true, frequently using counters for this condition.
- The modulus operator `%` extracts the last digit of an integer; floor division `//` discards the last digit.
- Tracing variable changes line-by-line is crucial to understand loop execution and debug logic.
- The multiplication table example ties counter behavior to practical output.
- Digit extraction logic prepares students for challenges involving number manipulation.
- Experimenting by writing these programs in an IDE consolidates understanding of loops and counters.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



While Loop

While Loop

- Counter

While Loop

- Counter
- Digits of a Number

While Loop

Counter

While Loop

Counter

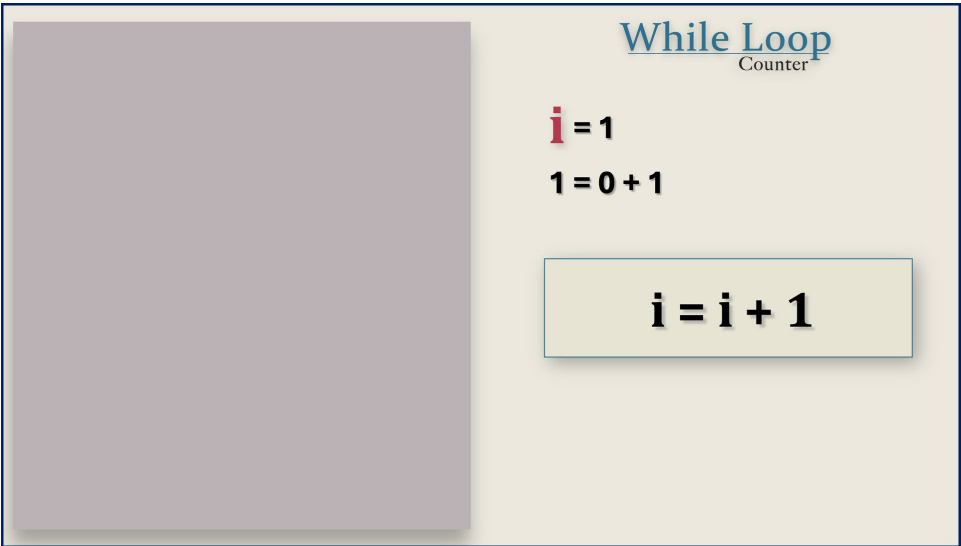
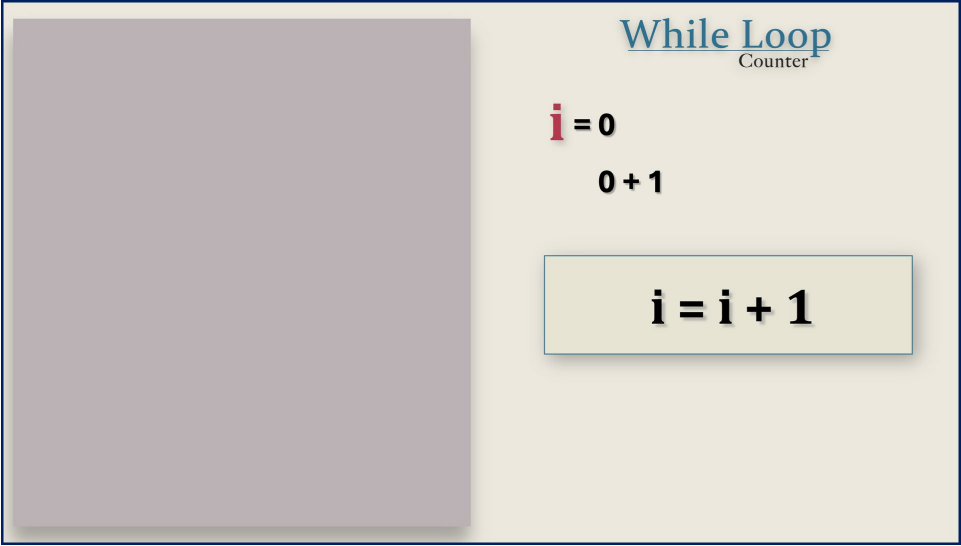
i = 0

While Loop

Counter

i = 0

i = i + 1



While Loop

Counter

i = 2

2 = 0 + 1 + 1

i = i + 1

While Loop

Counter

i = 3

3 = 0 + 1 + 1 + 1

i = i + 1

While Loop

Counter

i = 4

4 = 0 + 1 + 1 + 1 + 1

i = i + 1

While Loop

Example

i = 0

While Loop

Example

i = 0

i = i + 1

While Loop

Example

```
i = 0
while i < 10 :
    i = i + 1
```

While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
```

While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print(n * i)
```

While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print(n * i)
```



While Loop

Example

n = 3

i = 0

while i < 10 :

i = i + 1

print(n * i)



While Loop

Example

n = 3

i = 0

while i < 10 :

i = i + 1

print(n * i)

n

3



While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print(n * i)
```

i

0

n

3



While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print(n * i)
```

i

0

n

3



While Loop

Example

```
n = 3
```

```
i = 0
```

```
while i < 10 :
```

```
    i = i + 1
```

```
    print( n * i )
```

i

0

1

n

3



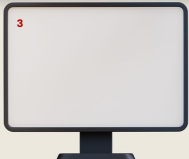
While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print(n * i)
```

i	n
0	3
1	

3



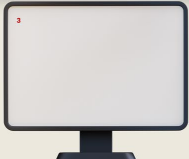
While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print(n * i)
```

i	n
0	3
1	

3



While Loop

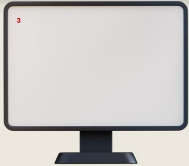
Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2

n
3

3



While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2

n
3

3
6



While Loop

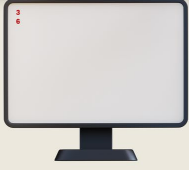
Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2

n
3

3
6



While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3

n
3

3
6



While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print(n * i)
```

i	n
0	3
1	
2	
3	

3

6

9

While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print(n * i)
```

i	n
0	3
1	
2	
3	

3

6

9

While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print( n * i )
```

i
0
1
2
3
4

n
3

```
3
6
9
```

While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print( n * i )
```

i
0
1
2
3
4

n
3

```
3
6
9
12
```

While Loop

Example

```
n = 3
```

```
i = 0
```

```
while i < 10 :
```

```
    i = i + 1
```

```
    print( n * i )
```

i

0

1

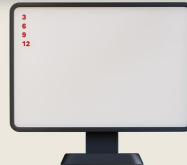
2

3

4

n

3



While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5

n
3

```
3
6
9
12
```

While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5

n
3

```
3
6
9
12
15
```


While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5

n
3

```
3
6
9
12
15
```

While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5
6

n
3

```
3
6
9
12
15
```

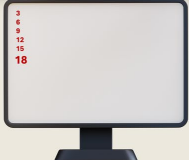
While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5
6

n
3



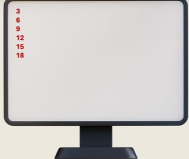
While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5
6

n
3



While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5
6
7

n
3

```
3
6
9
12
15
18
```

While Loop

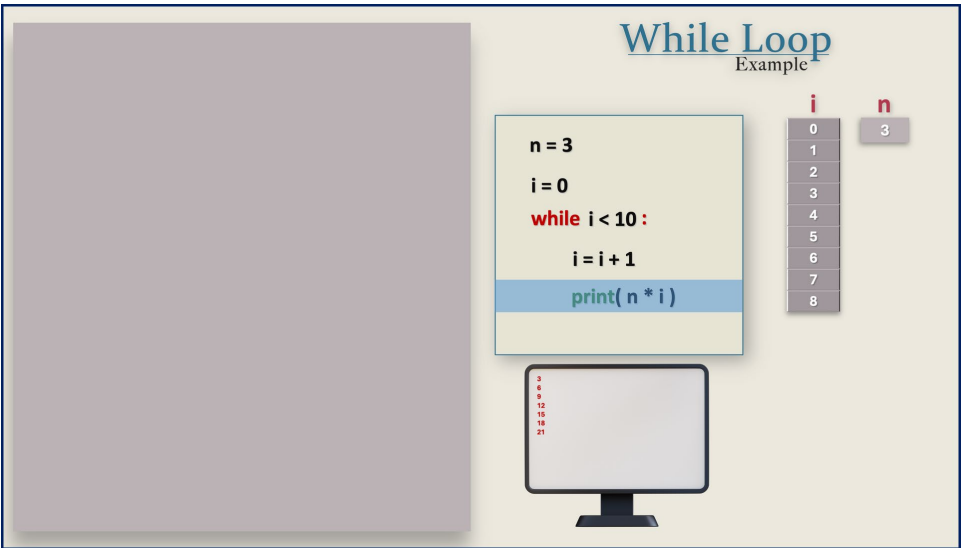
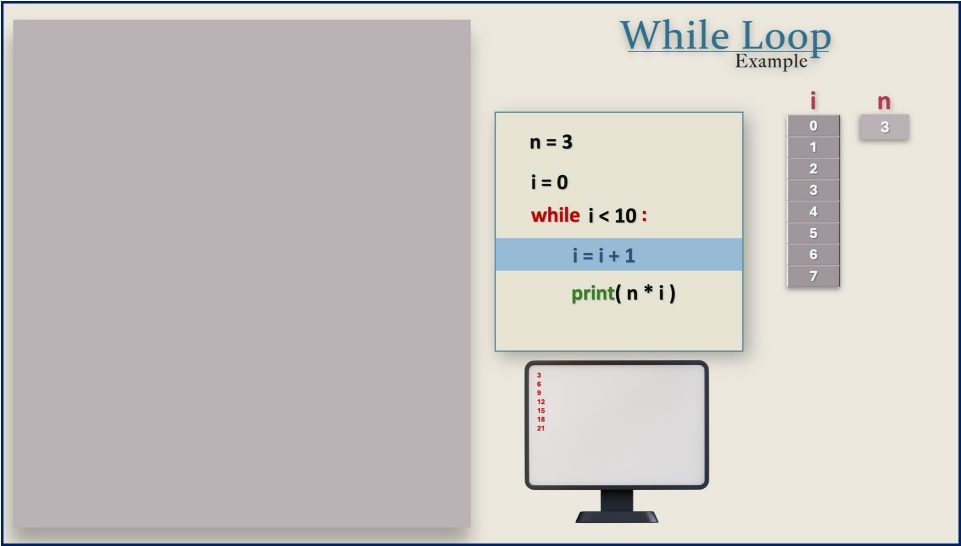
Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5
6
7

n
3

```
3
6
9
12
15
18
21
```



While Loop

Example

```
n = 3
```

```
i = 0
```

```
while i < 10 :
```

```
    i = i + 1
```

```
    print(n * i)
```

i

0
1
2
3
4
5
6
7
8

n

3

```
3
6
9
12
15
18
21
24
```

While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print(n * i)
```

i	n
0	3
1	
2	
3	
4	
5	
6	
7	
8	

3

6

9

12

15

18

21

24

While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print(n * i)
```

i	n
0	3
1	
2	
3	
4	
5	
6	
7	
8	
9	

3

6

9

12

15

18

21

24

While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5
6
7
8
9

n
3

3
6
9
12
15
18
21
24
27

While Loop

Example

```
n = 3
i = 0
while i < 10 :
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5
6
7
8
9

n
3

3
6
9
12
15
18
21
24
27

While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5
6
7
8
9
10

n3

3
6
9
12
15
18
21
24
27

While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5
6
7
8
9
10

n3

3
6
9
12
15
18
21
24
27
30

While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5
6
7
8
9
10

n
3

3
6
9
12
15
18
21
24
27
30

While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i
0
1
2
3
4
5
6
7
8
9
10

n
3

3
6
9
12
15
18
21
24
27
30

While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i

0
1
2
3
4
5
6
7
8
9
10

n

3

3
6
9
12
15
18
21
24
27
30

3
6
9
12
15
18
21
24
27
30

While Loop

Example

```
n = 3
i = 0
while i < 10:
    i = i + 1
    print(n * i)
```

i

0
1
2
3
4
5
6
7
8
9
10

n

3

3	X	1	=	3
3	X	2	=	6
3	X	3	=	9
3	X	4	=	12
3	X	5	=	15
3	X	6	=	18
3	X	7	=	21
3	X	8	=	24
3	X	9	=	27
3	X	10	=	30

3
6
9
12
15
18
21
24
27
30

While Loop

Example

```
3 X 1 = 3
3 X 2 = 6
3 X 3 = 9
3 X 4 = 12
3 X 5 = 15
3 X 6 = 18
3 X 7 = 21
3 X 8 = 24
3 X 9 = 27
3 X 10 = 30
```

```
n = 3
```

```
i = 0
```

```
while i < 10 :
```

```
    i = i + 1
```

```
    print(n, 'X', i, '=', n * i)
```

i	n
0	3
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	



While Loop

Example

While Loop

Example

- Digits of a Number

While Loop

Digits

n = 2546

While Loop

Digits

n = 2546



While Loop

Digits

n = 2546



2546

While Loop

Digits

n = 2546



2546 % 10

While Loop

Digits

n = 2546



6 = 2546 % 10

While Loop

Digits

n = 2546



6 = n % 10

While Loop

Digits

n = 2546



r = n % 10

While Loop

Digits

n = 2546



2546

r = n % 10

While Loop

Digits

n = 2546



2546 // 10

r = n % 10

While Loop

Digits

n = 2546



254 = 2546 // 10

r = n % 10

While Loop

Digits

n = 2546



254 = n // 10

r = n % 10

While Loop

Digits

n = 2546



n = n // 10

r = n % 10

While Loop

Digits

n = 2546



r = n % 10

n = n // 10

While Loop

Example

n = 4365

While Loop

Example

n = 4365

r = n % 10

While Loop

Example

```
n = 4365
```

```
r = n % 10
```

```
n = n // 10
```

While Loop

Example

```
n = 4365
```

```
r = n % 10
```

```
print(r)
```

```
n = n // 10
```

While Loop

Example

```
n = 4365
```

```
r = n % 10
```

```
print( r )
```

```
n = n // 10
```

While Loop

Example

```
n = 4365
```

```
while n > 0 :
```

```
    r = n % 10
```

```
    print( r )
```

```
    n = n // 10
```

While Loop

Example

```
n = 4365
```

```
while n > 0:
```

```
    r = n % 10
```

```
    print(r)
```

```
    n = n // 10
```



While Loop

Example

```
n = 4365
while n > 0:
    r = n % 10
    print(r)
    n = n // 10
```

r

While Loop

Example

```
n = 4365
while n > 0:
    r = n % 10
    print(r)
    n = n // 10
```

r

n

While Loop

Example

n = 4365

```
while n > 0:
    r = n % 10
    print(r)
    n = n // 10
```

r

n



While Loop

Example

n = 4365

```
while n > 0:
    r = n % 10
    print(r)
    n = n // 10
```

r

n

	4365



While Loop

Example

```
n = 4365
while n > 0:
    r = n % 10
    print(r)
    n = n // 10
```

r	n
--	4365



While Loop

Example

```
n = 4365
while n > 0:
    r = n % 10
    print(r)
    n = n // 10
```

r	n
--	4365
5	



While Loop

Example

```
n = 4365
while n > 0:
    r = n % 10
    print(r)
    n = n // 10
```

r	n
--	4365
5	

5

While Loop

Example

```
n = 4365
while n > 0:
    r = n % 10
    print(r)
    n = n // 10
```

r	n
--	4365
5	436

5

While Loop

Example

```
n = 4365
while n > 0:
    r = n % 10
    print(r)
    n = n // 10
```

r	n
--	4365
5	436
6	

5

While Loop

Example

```
n = 4365
while n > 0:
    r = n % 10
    print(r)
    n = n // 10
```

r	n
--	4365
5	436
6	43

5

6

While Loop

Example

n = 4365

while n > 0:

r = n % 10

print(r)

n = n // 10

r	n
--	4365
5	436
6	43
3	



While Loop

Example

n = 4365

while n > 0:

r = n % 10

print(r)

n = n // 10

r	n
--	4365
5	436
6	43
3	4



While Loop

Example

n = 4365

while n > 0:

r = n % 10

print(r)

n = n // 10

r	n
--	4365
5	436
6	43
3	4
4	



While Loop

Example

n = 4365

while n > 0 :

r = n % 10

print(r)

n = n // 10

r	n
--	4365
5	436
6	43
3	4
4	0

